

Question number	Answer	Additional guidance	Mark
3(a)(i)	An answer between 8 and 18 (hours) (1)		Graduate (1)

Question number	Answer	Additional guidance	Mark
3(a)(ii)	An answer showing the following steps: • correct values read from y axis and subtracted (1) • gradient calculated and appropriate units given (1)	2.8 - 1.5 = 1.3 1.3 ÷ 4 = 0.325 pmol dm⁻³ hour⁻¹ ALLOW 0.33 Conversion to other units will also need checking ALLOW 325 fmol in place of 0.325 pmol h⁻¹ / H⁻¹ for hour⁻¹	Expert (2)

Question number	Answer	Mark
3(b)	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material indicated as relevant. Additional content included in the response must be scientific and relevant. Indicative content Graph shows • as MDMA concentration in blood increases ADH concentration increases Table shows • after 9 hours of taking MDMA ADH levels still high / 3x that at 96 hours • as ADH concentration increases sodium ion concentration in the blood decreases • brain swelling is associated with lower sodium ion concentration in the blood From own knowledge ADH increases water reuptake by the kidney Deduction Increased water retention is to dilute the blood More water leaves blood by osmosis into brain tissue	Expert (6)

	Resulting in swelling of the brain	
--	------------------------------------	--

Level	Marks	Descriptor
	0	No awardable content.
1	1-2	Limited number of the most important or relevant scientific factors from the data/information provided are synthesised. No judgement is made.
2	3-4	Some of the most important or relevant scientific factors from the data/information provided are synthesised. A straightforward but accurate judgement is made.
3	5-6	Most of the important or relevant scientific factors from the data/information provided are synthesised. A detailed and accurate judgement is made.

